MAT2040 Linear Algebra Final Exam

SSE, CUHK(SZ)

25 July 2022

Seat No.: _____ Student ID: _____

i. The exam contains 10 questions. Answer all questions in this question book. Ask for extra answer sheets if needed.

ii. Without otherwise specified, all matrices and scalars in this paper are **real**.

iii. A table of notations is given in the second page, which you can checkout before the exam.

Question	Points	Score
1	5	
2	10	
3	10	
4	10	
5	10	
6	10	

Question	Points	Score
7	15	
8	10	
9	10	
10	10	
Total:	100	

This page has no questions, but a table of notations.

Table of Notations

$\mathbb{R}$	the set of real numbers Without otherwise specified, all matrices have entries from $\mathbb{R}$
$\mathbb{C}$	the set of complex numbers
$\mathbb{R}^n$	the set of all (column) vectors of n entries from $\mathbb{R}$ the n-dimensional Euclidean vector spaces
$\mathbb{C}^n$	the set of all (column) vectors of n entries from $\mathbb{C}$
$\mathbb{R}^{m\times n}$	the set of all matrices with entries from $\mathbb{R}$ and with size $m \times n$
$\overline{\lambda}$	complex conjugate. If $\lambda = a + bi$, where a and b are real, $\overline{\lambda} = a - bi$
$\overline{A}$	the complex conjugate of a matrix A, i.e., $\overline{A} = (\overline{a_{ij}})$ for $A = (a_{ij})$
$\mathbf{0}$	the zero vector or the all zero matrix, whose size is implied in the context or specified in the subscript
I_n	the $n \times n$ identity matrix
A^{T}	the transpose of matrix A
A^H	the conjugate transpose of a complex matrix A, i.e., $A^H = \overline{A}^{\mathrm{T}}$
$\det A$, $\det(A)$	the determinant of matrix A
Col$A$, Col(A)	the column space of matrix A
Null$A$, Null(A)	the null space of matrix A
adj$A$, adj(A)	the adjoint matrix of matrix A
$\langle \mathbf{x}, \mathbf{y} \rangle$	the inner product (scalar product) of vectors $\mathbf{x}$ and $\mathbf{y}$
$\|\mathbf{x}\|$	the norm of vector $\mathbf{x}$
$\dim V$, $\dim(V)$	the dimension of a vector space V
$V^\perp$	the orthogonal complement of subspace V
$[\mathbf{x}]_{\mathcal{B}}$	the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathcal{B}$
$C[a, b]$	the set of all the continuous functions defined on the closed interval $[a, b]$

1. (5 points) The real symmetric matrix A satisfies $A^2 + 6A + 8I = O$. Please show that $A + 3I$ is an orthogonal matrix.

2. Let $A = \begin{bmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. If $A\mathbf{x} = \mathbf{b}$ has more than one solution, answer the following questions:

   (a) (5 points) What is the value of a?

   (b) (5 points) Find an orthogonal matrix Q such that $Q^T AQ$ is a diagonal matrix.

3. $A = \begin{bmatrix} 2 & 2 & 0 \\ 8 & 2 & a \\ 0 & 0 & 6 \end{bmatrix}$ is similar to a diagnoal matrix Λ.

(a) (5 points) What is the value of a?

(b) (5 points) Find an invertible matrix P such that $P^{-1}AP = \Lambda$.

4. (10 points) Find an orthonormal basis for the solution set of the homogeneous linear equations $\begin{cases} 2x_1 + x_2 - x_3 + x_4 - 3x_5 &= 0 \\ x_1 + x_2 - x_3 + x_5 &= 0 \end{cases}$

5. (10 points) Suppose that a square matrix A has linearly independent columns.

   (a) Find the best least squares solution $\hat{\mathbf{x}}$ to $A\mathbf{x} = \mathbf{b}$. By taking those steps, give a formula (letters not numbers) for $\hat{\mathbf{x}}$ and a formula for the vector $\mathbf{p} = A\hat{\mathbf{x}}$.

   (b) Vector $\mathbf{p}$ is in which fundamental subspaces associated with A? Vector $\mathbf{r} = \mathbf{b} - \mathbf{p}$ is in which fundamental subspace? (Fundamental subspaces mean row space, column space, etc.)

   (c) Find the projection of a vector $\mathbf{x} \in \mathbb{R}^4$ onto the column space of B:

$$B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \\ 0 & -1 \\ 0 & -3 \end{bmatrix}$$

6. (10 points) Given the condition of matrix $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is full rank, what is the relationship between the line $l_1 : \frac{x-a_3}{a_1-a_2} = \frac{y-b_3}{b_1-b_2} = \frac{z-c_3}{c_1-c_2}$ and the line $l_2 : \frac{x-a_1}{a_2-a_3} = \frac{y-b_1}{b_2-b_3} = \frac{z-c_1}{c_2-c_3}$?

7. Every January, the product line accounts the the number of the skilled and the unskilled labors respectively, and transfers $\frac{1}{6}$ skilled workers to other apartments. Newly recruited unskilled labors make up the vacancy of transferred labors. $\frac{2}{5}$ of the new and old unskilled labors turn to the skilled labors after the whole year's training. Assume that at the n-th year's January, the percentage of the skilled and unskilled labors is x_n and y_n, denoted as vector $\begin{bmatrix} x_n \\ y_n \end{bmatrix}$.

   (a) (5 points) What is the relationship between $\begin{bmatrix} x_n \\ y_n \end{bmatrix}$ and $\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix}$? Written in the matrix form of $\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = A \begin{bmatrix} x_n \\ y_n \end{bmatrix}$;

   (b) (5 points) Compute the eigenvalues and the eigenvectors for matrix A;

   (c) (5 points) When $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$, find out $\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix}$.

8. (10 points) Find out the value of λ, which can make the system $\begin{cases} \lambda x_1 - x_2 - x_3 &= 1 \\ -x_1 + \lambda x_2 - x_3 &= -\lambda \\ -x_1 - x_2 + \lambda x_3 &= \lambda^2 \end{cases}$ has solution, and find out the solutions for the corresponding homogeneous system.

9. There are two homogeneous linear systems: system I is $\begin{cases} x_1 + x_2 &= 0 \\ x_2 - x_4 &= 0; \end{cases}$ and the solution set of system II is $k_1(0,1,1,0)^T + k_2(-1,2,2,1)^T$, $k_1, k_2 \in R$.

   (a) (5 points) Find out the homogeneous solution set of I.

   (b) (5 points) Do I and II have nonzero identical solutions? If nonzero identical solutions exist, find out all the nonzero identical solutions; If not, give the supporting explanations.

10. Calculate the determinants:

(a) (5 points) $\begin{bmatrix} a & 0 & 0 & \cdots & 0 & 1 \\ 0 & a & 0 & \cdots & 0 & 0 \\ 0 & 0 & a & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a & 0 \\ 1 & 0 & 0 & \cdots & 0 & a \end{bmatrix}$ (the size of this matrix is $n \times n$)

(b) (5 points) $\begin{bmatrix} x & -1 & 0 & \cdots & 0 & 0 \\ 0 & x & -1 & \cdots & 0 & 0 \\ 0 & 0 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x & -1 \\ a_n & a_{n-1} & a_{n-2} & \cdots & a_2 & x + a_1 \end{bmatrix}$